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Organ and Tissue Contribution to Metabolic Rate

M. Elia

Dunn Clinical Nutrition Centre, Cambridge, CB2 1QL, United Kingdom

The metabolic rate per kg body weight (or per kg fat-free mass) in young children may be two times greater than that in adults. Theoretically, this major decline with age could be due to reduction in the metabolic rate per gram of individual tissue or to a change in the proportion of different tissues, each of which has a different but relatively fixed metabolic rate. The available evidence largely supports the latter explanation.

The oxygen consumption of individual tissues in vivo can be estimated by making measurements of the arteriovenous concentration difference of oxygen across a tissue in conjunction with measurements of blood flow. Although the information obtained by this method is frequently approximate, it nevertheless provides important insights into the sites of thermogenesis (1). Adipose tissue, which comprises 21% body weight in the reference male and 33% in the reference female (2), has a very low metabolic rate *in vivo* (Kurpad and Elia et al., unpublished data and ref. 3—approximately 4.5 kcal/kg/day subcutaneous abdominal adipose tissue), which is very similar to that obtained in vitro (4). There are of course regional differences in the lipolytic activity and hormonal responsiveness of different adipose tissue depots, with the subcutaneous abdominal depot showing intermediate responsiveness between intra-abdominal and gluteal fat depots. Extrapolations of O_2 consumption from the subcutaneous abdominal fat depot to the adipose tissue of the whole body should be undertaken with caution. Nevertheless, such an undertaking suggests that adipose tissue (~15–20 kg) accounts for only about 4–5% of total O_2 consumption. Interestingly, extrapolations based on observations of the behavior of a variety of adipocytes in vitro (4) suggest a value of 4%, and a statistical approach based on measurements of body composition and whole body energy expenditure suggests a value of 5% for individuals with 20% body fat (5).

Although muscle is the largest tissue of the whole body, accounting for about 40% of adult body weight, its estimated resting metabolic rate is low (~10–15 kcal/kg/day), so its contribution to the total energy expenditure of the body is about 20% to 25%. These values are approximate and may be even lower than generally thought because estimates of forearm (Elia and Kurpad, unpublished data) and calf muscle

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blood flow (6) obtained directly by the xenon clearance technique have been found to be considerably lower than those obtained simultaneously using limb plethysmography. The majority of the resting energy expenditure of the body (~60%) arises from organs such as liver, kidney, heart, and brain, which account for only about 5% to 6% of body weight (Table 1). These tissues have a metabolic rate that is 15–40 times greater than an equivalent weight of muscle (using the more traditional value for the metabolic rate of muscle) and 50–100 times greater than adipose tissue (the variability is smaller when expressed in relation to the protein content of tissues (1). The lungs have not been included in Table 1 because of the lack of adequate data concerned with their oxygen consumption *in vivo*. Holliday et al. (12) estimated that the lungs contribute up to about 20% of the O_2 consumption of the body, which would imply that the five major organs in combination (heart, lung, kidneys, brain, and liver) contribute up to about 80% of the total energy expenditure of the body. However, the estimate of lung metabolic rate should be regarded as tentative, because it appears to have been based on the rate of O_2 consumption of slices of lung *in vitro*. This rate of O_2 consumption is considerably greater than that of lung tissue in a perfusion system (14).

The metabolic rate per gram of each organ appears to change little during growth and development (13). For example, Kennedy and Sokoloff (15) found no change in O_2 consumption/100 g brain/min between children aged 3 to 11 years. The values were also similar (although slightly higher) to those of adults. The higher metabolic rate of infants and children compared to adults (expressed in relation to body weight or fat-free mass or surface area) can therefore be largely explained by the presence of a larger proportion of metabolically active tissues in younger individuals (Table 2). The brain, which accounts for almost one-half of the total energy expenditure of the infant, decreases in proportion to body weight during growth and development so that in the adult it is responsible for only 20% of total energy expenditure. In contrast, skeletal muscle, which is metabolically less active than the brain and other organs, increases in proportion to body weight during growth and development. The contributions of different tissues to body weight and to total energy expenditure in the reference male and female and in a 7.5 kg infant are shown in Table 1.

Table 1 also illustrates that although the reference female has a lower metabolic rate per kg body weight, the discrepancy largely disappears when metabolic rate is expressed in relation to fat-free mass (see table footnote). The lower contribution of muscle to total energy expenditure in females is largely balanced by a greater contribution of other organs.

Using the changing patterns of organ size during growth and development (Table 2), it is possible to construct a figure which readily illustrates the contributions of different organs to total energy expenditure (Fig. 1). Interestingly, if no assumptions are made about organ metabolic rates, it can be shown that BMR per kg organ weights (sum of liver, kidney, brain, and heart) remains remarkably stable during growth and development, in contrast to the changes in metabolic rate per kg body weight or per square meter (Fig. 2).

TABLE 1. *Contribution of different organs and tissues to body weight and basal metabolic rate in the reference male, reference female, and a child*

	Tissue or organ weight (kg)[a]			Tissue or organ weight (% body weight)			Organ metabolic rate (kcal/kg/day)	Metabolic rate (% total)[b]		
	Reference male	Reference female	Child (0.5 y)	Reference male	Reference female	Child (0.5 y)		Reference male	Reference female	Child (0.5 y)
Liver[c]	1.8	1.4	0.263	2.57	2.41	3.51	200	21	21	14
Brain	1.4	1.2	0.713	2.00	2.07	9.51	240	20	21	44
Heart	0.33	0.24	0.040	0.47	0.41	0.53	440	9	8	4
Kidneys	0.31	0.275	0.053	0.44	0.47	0.71	440	8	9	6
Muscle	28.00	17.00	1.875	40.00	29.31	25.00	13	22	16	6
Adipose tissue	15.00	19.00	1.50	21.43	32.75	20.00	4.5	4	6	2
Miscellaneous tissues by difference (e.g., bone, skin, intestines, glands)	23.16	18.885	3.056	33.09	32.58	40.74	12	16	19	24
Total	70.00	58	7.5	100	100	100		100 (1680 kcal/day)	100 (1340 kcal/day)	100 (390 kcal/day)

[a] Tissue and organ weights for the reference male and reference female were taken from reference 2. Those for the 0.5-year-old child are consistent with results given by various authors (2, 7–9).

[b] The metabolic rates for the reference male (1,680 kcal/day) and reference female (1,340 kcal/day) are close to the mean values obtained by various standards for 30-year-old adults with a body mass index of 22.5 kg/m^2. The metabolic rate for the 0.5-year-old child (390 kcal/day) is close to the mean value timed by various standards for a male child with a weight of 7.5 kg and height of 0.65 m. If metabolic rate is expressed in relation to body weight, the values are 24.0, 23.3, and 52.0 kcal/day for reference male, reference female, and child, respectively. If metabolic rate is expressed in relation to fat-free mass (56.5 kg for the reference male, 42.0 kg for reference female—see reference man, ref. 2—and 6.1 kg for child) the values are 29.7, 31.9, and 63.9 kcal/kg fat-free mass/day, respectively.

[c] Several workers have estimated that the "liver" is responsible for 25–32% of adult BMR (see 10–13). These values are probably overestimates, since they were usually calculated from the O_2 consumption of liver plus portal drained viscera.

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TABLE 2. *Percentage contribution of various organs to the body weight of man*[a]

Age (y)	Brain	Liver	Kidneys	Heart	Brain, Liver, Kidneys, Heart	Muscle
Birth	12.2	4.53	0.82	0.68	18.2	21.3
1–5	8.3	3.45	0.54	0.56	12.9	—
6–10	6.7	3.05	0.54	0.60	10.9	—
11–15	3.7	2.63	0.45	0.51	7.3	36.2
16–20	2.6	2.55	0.46	0.54	6.2	—
21–30	2.2	2.54	0.45	0.54	5.7	45.2
31–40	2.2	2.48	0.46	0.54	5.7	41.8
41–50	2.3	2.56	0.45	0.58	5.9	40.2
51–60	2.4	2.42	0.47	0.56	5.9	—
61–70	2.7	2.14	0.41	0.63	5.9	33.9
>70	2.3	2.03	0.41	0.60	5.3	27.0

[a] Based on Korenchevsky (8). All the measurements were made in subjects who died of accidents, with the exception of the studies concerned with muscle, which included hospitalized patients. The values for muscle agree closely with those reported for normal growth and development by the other workers (2).

Because the organs of the body are so important in determining its metabolic rate, it is pertinent to consider whether variations in their size could explain differences in energy expenditure observed in pathophysiologic circumstances or even differences in energy expenditure between species.

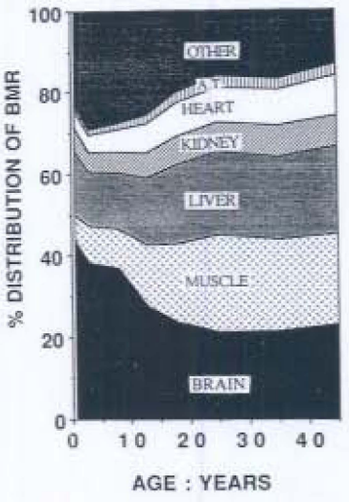


FIG. 1. The contribution of various organs and tissues to body weight and basal metabolic rate of a male during growth and development (0.5–45 years of age). The values for organ (tissue) weights for subjects between 3 and 45 years old are taken from Korenchevsky (1961) (Table 2) and for a 0.5-year-old baby (7.5 kg) from Table 1. Organ (tissue) metabolic rate is calculated from the product of organ (tissue) weight and organ/tissue metabolic rate. The adults were assumed to have a body mass index of 22.5 kg/m^2, height of 1.70 m, and weight of 65 kg. The children (0.5, 3, 8, 13 years old) were assumed to have heights (0.65, 0.97, 1.18, and 1.54 meters, respectively) and weights (7.5, 15, 22, and 45 kg, respectively) close to the 50th percentiles of the NCHS standards and those given by Frisancho (16). The basal metabolic rates for males aged 0.5, 3, 5, 8, 13, 18, 25, 35, and 45 years were taken to be 390, 780, 960, 1440, 1,700, 1,520, 1,590, and 1,550 kcal/day. For adults (18–45 years) the values are close to the mean values obtained from the various reference standards. For children (0.5–13 years) the values are also close to the mean values obtained from various reference standards (although not all standards cover the entire age range).

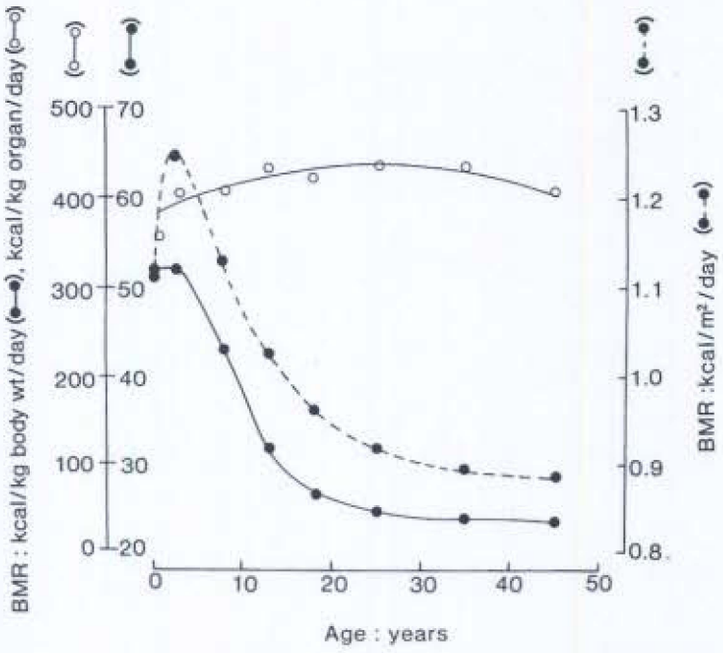


FIG. 2. Change in basal metabolic rate (BMR) during growth and development (0.5 to 45 years). BMR is expressed in relation to body weight (●——●), organ weight (○——○), and surface area (●--●). The graph was constructed from (A) the mean values of BMR, which were obtained using several standards; (B) organ weights, which were calculated using data from Korenchevsky (8) (Table 2) and from the values given in Table 1 for a 0.5-year-old child; and (C) surface area, which was calculated using the formula of Dubois and Dubois (Table 1 in Elia, Energy Expenditure in the Whole Body). The transient rise in BMR/m^2 between 0.5 and 3.5 years of age is not an artifact but a well-known developmental change.

ORGAN SIZE IN MAN AND OTHER MAMMALS

The metabolic rate/kg body weight of mammals decreases as the animal gets bigger. In this respect, mice have a metabolic rate that is five times greater than that of humans and ten times greater than that of elephants (17). Over a wide range of mammalian species a number of workers have found that resting energy expenditure is proportional to weight$^{0.75}$ (i.e., for every percent increase in weight, there is approximately 0.75% increase in metabolic rate). The mouse, the elephant, and many other mammals conform approximately to this general relationship.

It is possible that the lower metabolic rate per kg body weight of larger animals is due largely to the relatively smaller size of metabolically active tissues (i.e., organ size). If the metabolic rate per gram of individual organs varied little among different mammals, then organ size would be directly related to the metabolic rate of the mammal and to body weight$^{0.75}$.

I have attempted to examine such relationships using values of organ weights

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TABLE 3. *Relation of organ weight to body weight (organ weight = K × body weight^x) in approximately mature mammals*[a]

	Live weight (g)	Liver (g)	Kidneys (g)	Brain (g)	Heart (g)	Lung (g)	All 5 organs (g)	All orga minus lung (g)
Elephant	6,650,000	6,300	1,200	5,700	2,200	(6,650)	22,050	15,400
Steer	700,000	5,000	1,000	500	2,300	3,900	12,700	8,800
Horse	600,000	6,700	1,600	670	4,250	5,400	13,220	18,620
Dairy cow	488,000	6,460	1,160	400	1,880	3,600	13,440	9,840
Hog	125,000	1,600	260	120	350	1,300	3,530	2,230
Human	60,000	1,700	250	1,300	320	1,000	4,570	3,570
Sheep	52,000	960	160	106	280	710	2,216	1,506
Dog	10,000	420	70	75	85	120	770	650
Monkey	4,500	110	21	42	23	30	226	196
Guinea pig	800	27	5.6	4.7	2.3	5.0	44.6	39.(
Rat	250	12	2.1	2.0	0.94	1.3	18.34	17.(
$KW^{x,b}$		$0.39W^{0.70}$	$0.066W^{0.71}$	$0.76W^{0.71}$	$0.03W^{0.88}$	$0.016W^{0.92}$	$0.406W^{0.77}$	$0.448W^{0}$
Correlation coefficient		0.97	0.97	0.94	0.96	0.97	0.97	0.(

[a] Calculated using data from Brody (17).
[b] *K*, constant; W, body weight in grams; *x*, exponent.

provided by Brody (17) (Table 3). A less complete analysis has been undertaker previously (10,12). For abdominal organs (liver and kidney) and brain it would appea that organ size is approximately proportional to body weight$^{0.75}$, but for the thoracic organs (heart and lung), the exponent is higher. The weight of all the major organs combined (liver, kidney, lung, brain, and heart) is proportional to body weight$^{0.77}$ which implies that the overall differences in metabolic rate between the mammals examined can be largely explained by a change in organ size. However, it should be noted that Brody (17) also examined the relationship between metabolic rate and body weight in an unknown number and type of mature mammalian species and obtained generally higher exponents (kidney, 0.846; liver, 0.867; brain, 0.697; and heart, 0.984) than those indicated in Table 3. These other exponents imply that as mammals get bigger, there is not only a smaller contribution of organ weight to body weight but also a reduction in metabolic rate per gram of organ.

There is also some uncertainty as to whether the metabolic rate per unit organ weight *in vivo* is fixed among different mammals. Schmidt et al. (18) found that when the brain metabolic rate is expressed in ml O_2/g brain/min, the values for small spider and rhesus monkeys weighing 2.8 kg to 5.8 kg are similar to that of the human brain (see also ref. 19). Approximately similar values have also been found for rabbit and dog brain (cited in ref. 18). On the other hand, more recent work (20) suggests that the metabolic rate of the rat cerebral tissue is two times greater than that of the human, and it was suggested that this might be due to the presence of a much smaller proportion of supporting glial cells (presumably lower metabolic rate) in the rat brain than in the human brain. Discrepancies can also be found in the metabolic rates o

other mammalian organs in vivo, but these are probably at least partly due to the use of different techniques for assessing blood flow in animals under different conditions (e.g., fed or fasted) and the assumptions used to calculate oxygen consumption per gram of tissue (e.g., uncertainties about tissue or organ size in animals that are not sacrificed and uncertainties about the estimated blood flow and the supply of blood in tissues that have dual sources). The work in vitro is also conflicting (e.g., see refs. 14,21,22), and the measurements often do not reflect those in vivo in any systematic way (18). Indeed, the metabolic rate of tissues in vitro can be altered by changing the composition of the medium in which they are immersed. Nevertheless, there is no doubt that differences in organ size between species, with or without some variation in metabolic activity, can go a long way toward explaining the differences in metabolic rates between different mammals.

One other observation that may be of interest to human energy metabolism concerns the ratio of organ weights to body weights in mature mammals of the same species. This ratio appears to get smaller with larger animals (17). If the same is also the case in humans, it could provide an explanation for the observation of several workers that larger individuals have a lower metabolic rate per kg body weight (or per kg fat-free mass) than smaller individuals (see chapter by Elia, *Energy Expenditure in the Whole Body*).

CHANGES IN ORGAN SIZE AND METABOLIC RATE IN PATHOPHYSIOLOGIC STATES

Koong and colleagues (23,24) measured organ weights and metabolic rate in groups of pigs with different nutritional backgrounds. One group lost weight initially and then compensated by rapid weight gain, another achieved rapid weight gain initially and lost weight later, and a third gained weight steadily. At the end of 10 weeks the three groups of animals had achieved similar weights, but their metabolic rate differed by 40%. The animals that had the most recent weight gain had the highest fasting metabolic rate and heaviest weight of liver, kidney, pancreas, and intestine. These weights were 25% to 50% greater than in the animals that had lost weight recently and had the lowest metabolic rate. Similar observations were made in lambs. Unfortunately, it is not possible to assess from these reports whether there were differences in muscle mass, degree of adiposity, or metabolic rate per kg fat-free mass.

Other experiments (25) confirmed that the splanchnic tissues of lambs account for approximately one-half the whole body O_2 consumption, as in cattle (26), and went on to show that in lambs given a maintenance diet, the contribution of portal drained viscera to whole body O_2 consumption was reduced by about one-third compared to animals that fed ad lib and gained weight. The contribution of the liver to whole-body O_2 consumption in the lambs given a maintenance diet was reduced by about one-half compared to the control animals, but this was not considered to be associated with a change in blood flow or O_2 consumption per g liver. This suggests that the diet had an effect on the size of the liver but little or no effect on its overall metabolic activity.

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In studies (27–29) carried out at the Dunn Nutrition Unit in Cambridge, England changes in liver size (and protein content) relative to muscle mass (gastrocnemius weight and protein content) were studied in rats that received diets deficient in either protein or energy. In general, rats given the energy-restricted diets maintained essentially a normal distribution of protein between liver and muscle. In contrast, in the rats given protein-deficient diets, there was a preservation or an increase in muscle mass at the same time as a major depletion in liver mass. The result was a twofold change in the ratio of liver to muscle protein. These last changes were also associated with the development of hypoalbuminemia, which was not observed in the rats given energy-restricted diets. The authors considered the effects to be relevant to the development of kwashiorkor and marasmus in humans.

In further studies in which corticosterone was given to protein deficient rats (29) there was a large and rapid increase in liver weight and protein content (twofold within four days), an associated increase in the plasma albumin concentration, and a decrease in the size of muscle. Amino acids made available from muscle protein catabolism were presumably used to build liver protein and synthesize albumin. No such changes were observed in control animals given saline. Although the dose of corticosterone used in these experiments was pharmacologic (circulating concentration increased tenfold) the study emphasizes the potential for major and rapid changes in the distribution of protein between different components of fat-free tissue in the rat.

In more recent studies in rats, Wusteman and Elia (30) showed that aseptic abscesses induced by subcutaneous injections of turpentine also produced shifts of protein from muscle to liver (Table 4). The protein content and size of the live

TABLE 4. *Effect of subcutaneous turpentine injection and pair feeding on the weights, protein content, and protein synthetic rate of the gastrocnemius muscle and liver of 40-day-old rats (48 h post injection)*[a]

	Control fed ad-lib	Turpentine injected	Pair-fed
Tissue mass			
Muscle (g)	1.45	1.06[c,f]	1.41
Liver (g)	6.28	6.38	4.67[c]
Protein content			
Muscle (mg)	259	206[c,g]	249
Liver (mg)	926	1014[c,f]	801
Protein synthesis			
Muscle (%/day)	16.3	8.4[d,f]	13.3[b]
Liver (%/day)	120	154[b,e]	119

[a] Items noted in [b],[c], and [d] are significantly different from control animals. Items noted in [e], [f], and [g] are significantly different from pair-fed animals. From Wusteman and Elia (30).
[b] $p < 0.05$
[c] $p < 0.01$
[d] $p < 0.001$
[e] $p < 0.05$
[f] $p < 0.01$
[g] $p < 0.001$

increased despite the negative nitrogen balance and anorexia, which caused loss of liver protein. These changes in the turpentine-injected animals were found to be related to a decrease in muscle protein synthesis and an increase in liver protein synthesis (30), as measured by the phyenylalanine flooding technique. None of these changes could have been predicted from measurements of fat-free mass, nitrogen balance, or whole-body protein turnover.

These observations suggest that the change in energy expenditure, which frequently occurs in sepsis and other hypermetabolic conditions, may be at least partly due to changes in the relative proportion of metabolically active tissues (such as liver) to less active tissues, such as muscle. Investigations to assess the relationships between organ size, body composition, and metabolic rate in humans with disease are currently in progress. Interestingly, a previous study reported that anorexia nervosa was associated with substantial loss of both muscle and visceral organ size as assessed by computerized axial tomography, but in patients with malignancy there was preferential preservation of organ size (31). No measurements of metabolic rate were reported in this study, so that it is not possible to relate variations in organ size to energy expenditure.

VARIABILITY OF RESTING ENERGY EXPENDITURE IN DISEASE

Figure 3 shows the results of 73 measurements of resting energy expenditure in a group of patients referred for nutritional support at Addenbrooke's Hospital, Cambridge (36 for sepsis; 7 for surgery; 4 for surgery plus sepsis; and 26 for no sepsis, no surgery—e.g., Crohn's disease) and of 92 measurements in a group of normal, healthy subjects (mean age 27 ± 8 years, weight 65.4 ± 10 kg, height 1.68 ± 0.07 m, BMI 23 ± 6 kg/m^2). The values are expressed as a percentage of the energy expenditure predicted by the Harris-Benedict standards for normal individuals with the same weight and height as those studied. Two important observations emerge. First, the scatter in results in the patients (Fig. 3—middle graph) is greater than those obtained in the group of normal individuals (Fig. 3—upper graph). Second, the values in the patients are skewed and shifted to the right, which implies that there was a tendency for this group of subjects to be hypermetabolic. Both the scatter and the systematic bias of the prediction is reduced by taking into account disease activity and malnutrition using a simple nomogram such as the one indicated in Fig. 4 (see also ref. 32). However, the scatter in results still remains substantially greater than in normal individuals.

The variable resting energy expenditure observed in clinical practice is almost certainly due to a variety of factors. Although disease tends to increase energy expenditure, the variation is large, even for patients suffering from the same disease. This is partly due to difficulties associated with accurately assessing the severity of diseases such as sepsis and partly to the changes in energy expenditure which occur with time (especially in those with complications). In addition, disease states are often associated with anorexia and malnutrition, which may affect energy expen-

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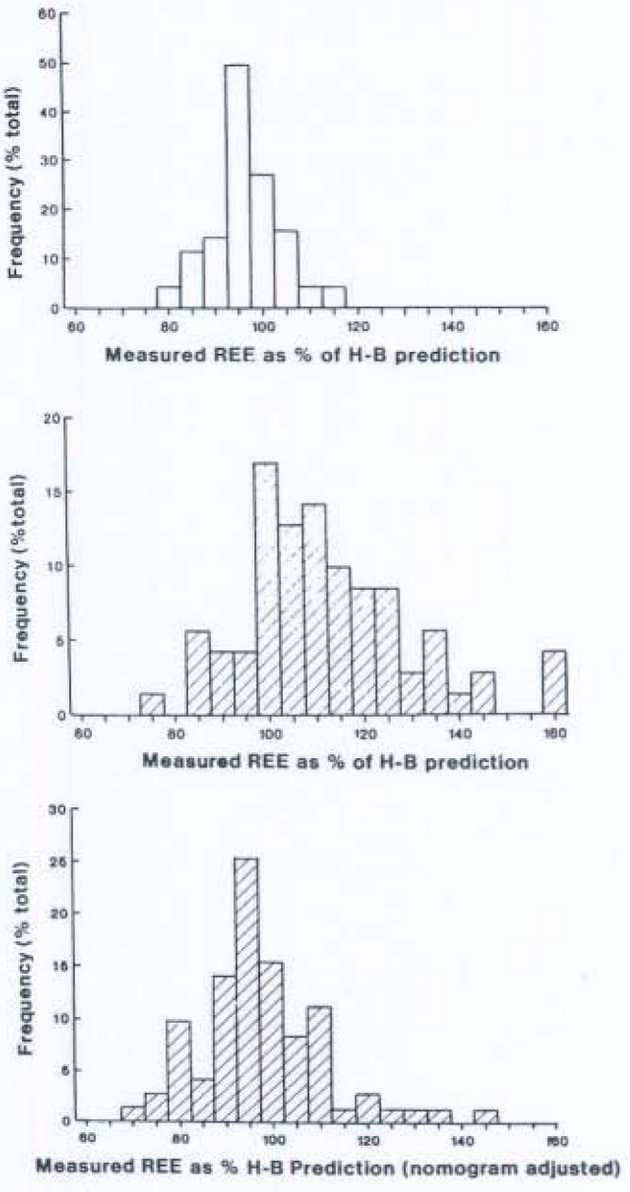


FIG. 3. Distribution of measured resting energy expenditure (REE) when expressed as a pe centage of Harris-Benedict (H-B) prediction (normal subjects and patients). **Upper graph**: Norm subjects (n = 92; body mass index, 18–30 kg/m^2). **Middle graph**: patients (n = 73) referred f nutritional support (see text). **Lower graph**: Measured REE in the patients shown in the midd graph expressed as a percent of Harris-Benedict prediction after adjustment of metabolic ra according to the nomogram illustrated in Fig. 4.

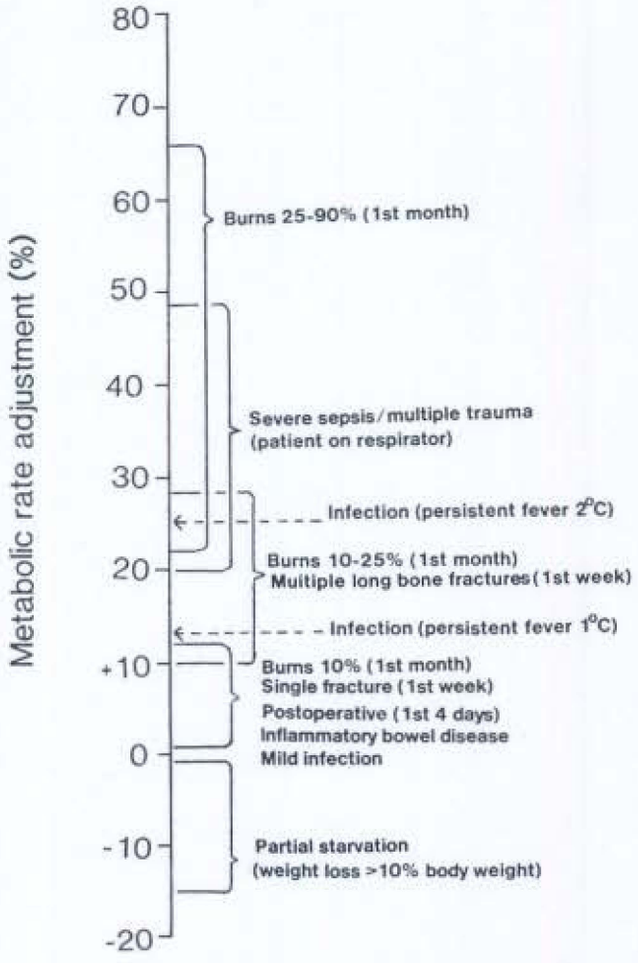


FIG. 4. The approximate effect of disease and malnutrition on basal metabolic rate. The change in metabolic rate is expressed as a percentage change (% adjustment).

diture in the opposite way to the disease. There are also the confounding effects of a variety of drugs and of blood transfusions, which may cause pyrexia and hypermetabolism, and the practice of nursing some patients, such as those in intensive care units, in a near naked state (at variable ambient temperatures) so that important clinical signs can be readily observed. There are also the effects of anxiety, abnormalities in fluid and electrolyte status, which affect body weight, and therefore the prediction of metabolic rate.

Pyrexia has also been used to predict metabolic rate (approximately 13% increase in metabolic rate per °C rise in temperature), but the variability is large, and fluctuations in temperature are also often large and variable.

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O_2 CONSUMPTION OF INDIVIDUAL TISSUES IN DISEASE STATES

Although the causes of abnormal energy metabolism in disease states are likely to be multiple, the earlier discussion raises the possibility that abnormality might be due to a change in the proportion of metabolically active tissues (e.g., organs) to less active tissues (e.g., muscle). Such changes may be common in various clinical states, but the available information in humans is scant. It seems unbelievable that there is a substantial change in the size of the brain in acute systemic disease. In some situations it also seem unlikely that an increase in the size of other organs is the major cause of hypermetabolism. For example, major changes in energy expenditure can occur very rapidly in some patients (e.g., post-trauma patients). They can also occur within minutes when hormones such as catecholamines are given either alone or in combination with glucagon and corticosteroids so that they mimic the stress response. It is inconceivable that redistribution of fat-free tissue occurs so rapidly.

Another important consideration is the magnitude of the hypermetabolic response, which may be greater than 50% above normal. If this was due to an increase in organ size without a concomitant increase in energy expenditure per gram tissue, the organs would have to double in size. Such major organomegaly, which is readily detected clinically or on radiologic and scanning investigations, does not normally occur in patients with marked hypermetabolism. Therefore, the cause of hypermetabolism in these types of patients is likely to be due to a greater energy expenditure per g individual organs or tissues (with or without a concomitant small to moderate increase in organ size). This conclusion is consistent with experimental observations in burned patients, which show an apparent increase in O_2 consumption in a wide variety of tissues: peripheral tissues, the size of which can be readily measured, as well as splanchnic tissues, kidneys, and heart (33). The damaged skin is not the major site of increased energy expenditure, although the damage initiates a series of events that leads to increased energy expenditure in a variety of other tissues in the body.

There is very little information about organ metabolic rates in human malnutrition or starvation (i.e., O_2 consumption or energy expenditure per g individual tissues). It is therefore difficult to assess whether adaptation occurs as a result of a disproportionate decrease in the size of metabolically active tissues such as the splanchnic organs or a decrease in metabolic rate per g individual tissue (34). In experimental rats, total starvation for 12 days has been reported to lead to a much greater fractional loss of organs such as liver and kidney than skeletal muscle (35). This could be an important explanation for the reduction in metabolic rate observed in starvation.

ENERGY COST OF SPECIFIC METABOLIC PROCESSES IN HEALTH AND DISEASE

At present there is fragmentary information about the metabolic processes responsible for the energy expenditure of individual tissues or the whole body in health

or disease. The energy expenditure associated with protein synthesis; various substrate cycles; Na-K exchange across cell membranes, which alone is estimated to account for about 20% of BMR in a variety of mammals (36); and brown fat activity have all been implicated as being important in normal individuals, although their importance may vary with age (e.g., brown fat activity is probably less important in adults than in infants). Because the activity of several of these processes may be markedly increased in disease states, they may contribute significantly to the hypermetabolism of individual tissues.

One of the complexities of metabolic pathways or cycles is the fact that they may not simply occur in one tissue but span across two or more tissues. For example, glycolytic products such as lactate pyruvate and alanine, which are released by peripheral tissues, may be used by the liver to produce glucose. This may then be reutilized by the peripheral tissues to form more glycolytic products. Similarly, the triglyceride fatty acid cycle involves the hydrolysis of triglyceride in adipose tissue to release fatty acids and glycerol. At the same time, fatty acids may be used to synthesize triglycerides by esterification of glycerophosphate. Because adipose tissue lacks significant activity of glycerokinase, the glycerol released from the hydrolysis of triglyceride cannot be utilized directly. Instead, it is released into the circulation and taken up by the liver, where it is converted to glucose. The glucose finally returns to adipose tissue, where it is used to form glycerophosphate, which allows the triglyceride fatty acid cycle to be completed.

The energy cost of these and other cycles can be estimated using a stoichiometric approach. The ATP required in a substrate cycle (assessed from classic metabolic pathways) is multiplied by the caloric equivalent of ATP. The value for this equivalent may vary somewhat, depending on the degree of coupling of oxidative phosphorylation in the mitochondria and the type of fuel oxidized (37–39). For example, if the combustion of starch yields 679 kcal (2,840 kJ)/glucosyl residue, and its oxidation within the body results in a net gain of about 36.7 ATP/glycosyl residue, the caloric equivalent of ATP is 18.5 kcal/mole ATP (679/36.7). For the same degree of mitochondrial coupling, the corresponding value for fat is close to 18.9/kJ mole ATP. For amino acids (protein) it is about 20.8 kJ/mole ATP, and for glycogen it is 18.0 kJ/mole ATP (37,39). In the calculations used in this chapter, the caloric equivalent of ATP is taken to be 19 kcal/mole (about 80 kJ/mole).

If the activity of a substrate cycle and its ATP requirement are known, then the energy cost of the cycle can be readily estimated and expressed in relation to the energy expenditure of the body. However, the ATP requirement of different cycles varies (e.g., 8 ATP for triglyceride–fatty acid recycling and 1 ATP for glucose-glucose 6-P recycling), and in some cases there is still some uncertainty about the precise requirement. For example, in the protein–amino acid cycle (protein turnover), 2 ATP are required for the activation of amino acids to amino acyl tRNA and a further 2 ATP for elongation (peptide bond synthesis), making a total requirement of 4 ATP per mole amino acid incorporated into protein. However, this requirement assumes that activated amino acids are incorporated into protein with 100% efficiency. If some amino acyl tRNA is hydrolyzed without incorporation of the amino acid into

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TABLE 5. *The activity and energy cost of some substrate cycles in normal and burned subjects*

Substrate cycle	Mole ATP used/mole substrate cycled[a]	Normal subjects: Rate of substrate cyling mole/70 kg/day[a]	Normal subjects: Energy cost of cycle kcal/d[b]	Normal subjects: Energy cost of cycle %BMR[c]	Burned subjects: Rate of substrate cycling mole/70 kg/day	Burned subjects: Energy cost of cycle kcal/d	Burned subjects: Energy cost of cycle %BMR	Δ Energy cost of cycle × 100[d] ΔBMR	Reference for rates of substrate cycling
Protein-amino acid (protein turnover)	(4)	3.1	236	14	6.6	502	15–20	19–32	(43)[e]
Glucose-"lactate"	4	0.42	32	1.9	1.6	122	3.7–4.9	6.5–11	(33)[f]
		0.24	18	1.1	—	—	—	—	(44)
		0.21	16	1.0	—	—	—	—	(45)
Triglyceride-fatty acid	8	0.12	18	0.8	0.53	81	2.9	12	(46)[g]
		0.04	6	0.36	—	—	—	—	(47)
		0.13	20	1.3	—	—	—	—	(48)
Glucose-glucose 1P and Fructose 6P-	1	0.4	8	0.33	0.96	18	0.65	2	(46)[g]
		0.29	6	0.33	—	—	—	—	(49)
Fructose 1-6P		0.69	13	0.85	—	—	—	—	(50)

[a] In the case of the protein-amino acid cycle, this refers to the molar rate of amino acid cycling. For the glucose-lactate cycle the molar rate refers to glucose cycling (not lactate), and for the triglyceride fatty acid cycle it refers to the triglycereide (not fatty acid).

[b] Energy cost (kcal/day) = substrate cycling rate (moles/day) × ATP used/mole substrate cycled × 19 (kcal/mole ATP).

[c] When basal metabolic rate measurements in normal adults were not available, it was assumed that they expended 1,650 kcal/day (70 kg body weight).

[d] Δenergy cost of cycle (kcal) = energy cost of cycling in burned subjects − energy cost of cycling in normal subjects. ΔBMR = BMR of burned patients (70 kg) − BMR of normal subjects (70 kg).

[e] The burned subjects in this study (mean burn size − 65% of total surface area) were studied at 11 ± 4 days. The range in values for %BMR and for Δenergy cost cycle/ΔBMR in the burned subjects were calculated assuming metabolic rate was elevated 50–85% above normal.

[f] The range in values for %BMR and 100 × Δenergy cost cycle/ΔBMR in the burned subjects was calculated assuming metabolic rate was 50–85% above normal. The estimated rate of cycling includes cycling of lactate and other 3-carbon glycolytic products, e.g., alanine (in normal man, glucose-alanine cycling occurs at about one-half the rate of glucose-lactate cycling—ref 45).

[g] Both the control and burned patients in this study were a combination of adults and children. The burned subjects (74 ± 3% surface area burn) were studied 20 ± 5 days after injiury when their metabolic rate was found to be 23% above normal (estimated to be ~40 kcal/kg/day compared with ~32.5 kcal/kg/day for the control subjects).

protein, the ATP requirement per peptide bond synthesis will be greater than 4 moles. Furthermore, the ATP required to maintain the turnover of RNA (mRNA, tRNA, rRNA—refs. 40,41) and transport of amino acids is not included in the preceding estimate of ATP requirement for protein synthesis; nor is the one ATP required to initiate protein synthesis and another ATP required to initiate nonlysosomal protein breakdown. Although the energy cost of some of these processes is probably small compared to the energy cost of protein synthesis per se, the value of 4 ATP/mole peptide bond synthesized should be nevertheless regarded as a minimum value (see also ref. 42).

Table 5 illustrates the activity of three cycles: glucose lactate recycling, protein amino acid recycling (protein turnover and triglyceride/fatty acid recycling), and the associated energy expenditure calculated in absolute terms and as a percentage of resting energy expenditure in both normal subjects and those with a severe form of injury (burn injury). Although the majority of basal energy expenditure remains unaccounted for, the increased activity of these three cycles in burned patients accounts for up to one-half of the rise in energy expenditure (Table 5). A further contribution arises from the increased energy cost associated with cardiac output, which may increase more than two times in burned patients. The energy expenditure of the heart may double in burn patients (33), and there may be an increase in the work of breathing, which frequently produces a twofold to fourfold increase in minute ventilation.

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DISCUSSION

Dr. Pi-Sunyer: I'd like to commend Dr. Elia for what I think is a very scholarly and thorough presentation of the problems related to using derived equations for calculating BMR metabolic rate in individuals.

To evaluate whether these derived equations are reasonable or not, we have to ask the question, what are we utilizing the derived equation for? In some cases the derived equation will be perfectly justified and in other conditions will be totally inappropriate.

It seems to me if one is treating a patient in a metabolic ward and trying to feed that patient, the use of an equation from Harris and Benedict to get an estimate of the amount of nutrient that is required, with all the manipulations that have to be done, the derivation of the equation is incredibly inaccurate. However, such equations seem to me perfectly reasonable, because they can be corrected relatively quickly as the metabolic state of the patient is being watched carefully in a metabolic ward.

On the other hand, if one is to derive an equation to diagnose an abnormal metabolic state, the danger of giving an incorrect diagnosis to an individual is rather high.

If the calculations are done for a public health reason, it seems to me since one is going to want to err on the high side anyway, using an equation like the Harris-Benedict equation that might be somewhat high may not necessarily be inappropriate to calculate the public health needs of a population group.

The problems become very great if one is to use an equation to try to characterize a particular population group—for instance, the Pima Indians or the black population of America or the Mexican Indian population—to measure their metabolic rate, compare it to a derived equation, and state, "This particular group is hypometabolic." I think that since we are dealing with uncertainty, it depends a great deal how we're to utilize the result of that calculation: for an individual or for a group of individuals.

Dr. Elia comments on the utilization of surface areas versus the utilization of fat-free mass. He seems rather unhappy with the use of surface area, feeling that it is a rather inaccurate way of trying to derive whether an individual is hypometabolic or hypermetabolic.

It's my impression that surface area is likely to be more accurate than many ways that are being utilized for trying to derive fat-free mass of individuals. We have an enormous number of commercial companies and individuals selling machines that are supposed to measure lean body mass accurately and actually do a rather inaccurate job of it. And yet those derived fat-free mass figures are being used as denominators for calculating metabolic

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